

# Lab 4: IMU Odometry

## Dead Reckoning using Inertial Measurements

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### Assignment Overview

This lab focuses on dead reckoning using IMU data, including magnetometer calibration, gyroscope integration for orientation estimation, and accelerometer integration for velocity and position reconstruction. Data was collected for two motion patterns: circular walking and square walking.

## 1 Circle Walking Data

### 1.1 Magnetometer Calibration

**Task:** Plot the N vs. E components of magnetic field and apply calibration to your dataset. Plot two marker styles on the same figure to display the data before and after calibration (Fig. 1).

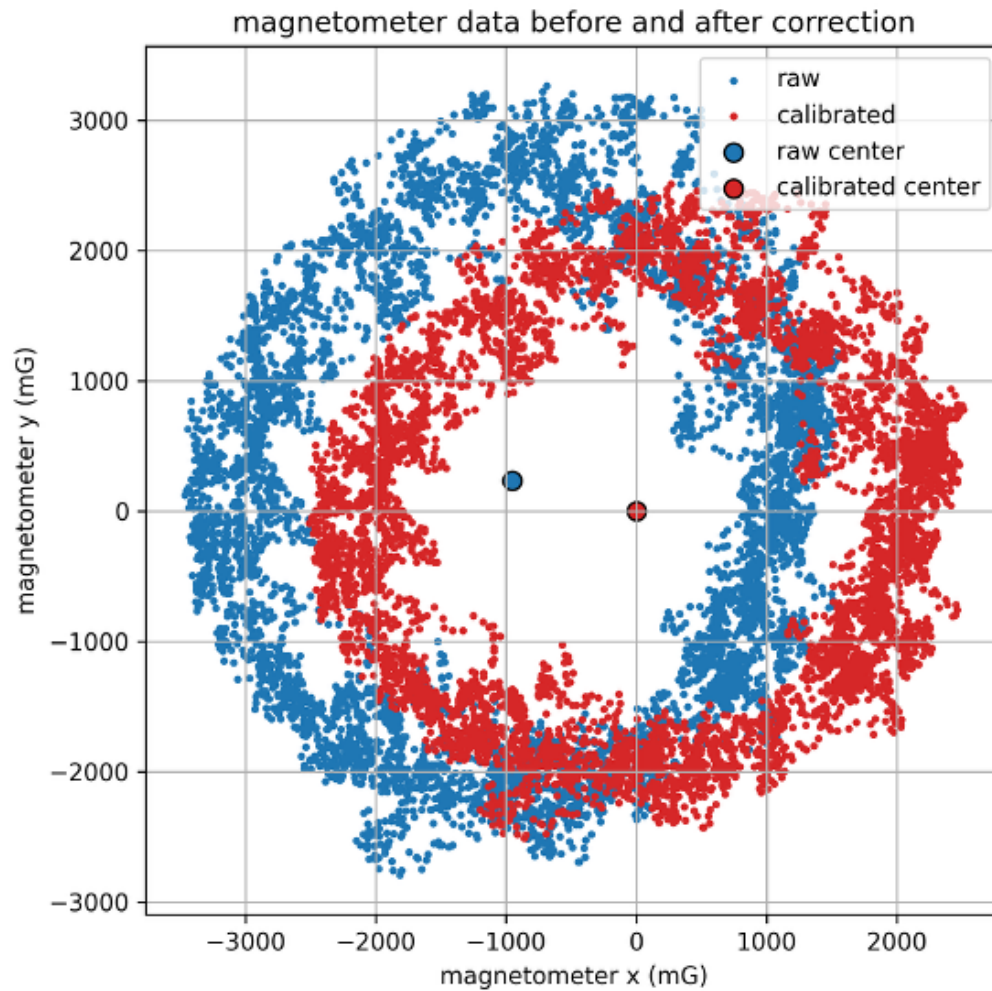


Figure 1: Magnetometer calibration for circle walking - before and after calibration

## 1.2 Gyroscope X-axis Integration

**Task:** Plot rotational rate around X and rotation from the gyro around X (integrate once) on two subplots (Fig. 2).

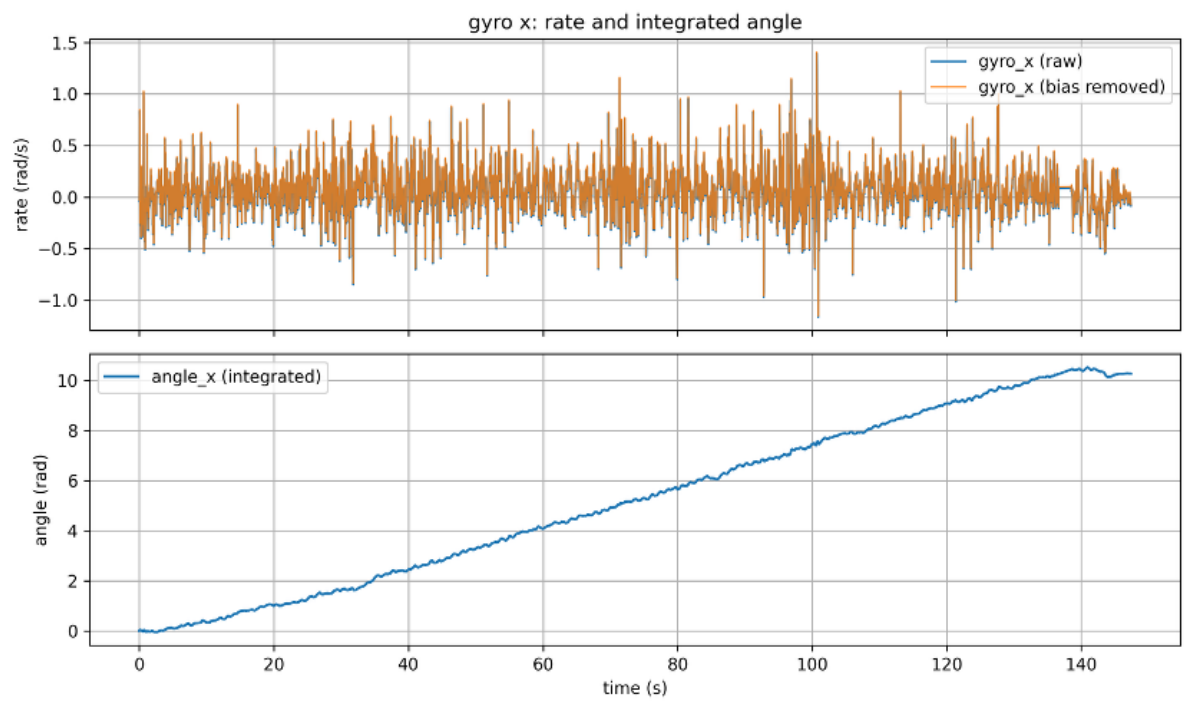


Figure 2: Gyroscope X-axis: rotational rate and integrated rotation

### 1.3 Gyroscope Y-axis Integration

**Task:** Plot rotational rate around Y and rotation from the gyro around Y (integrate once) on two subplots (Fig. 3).

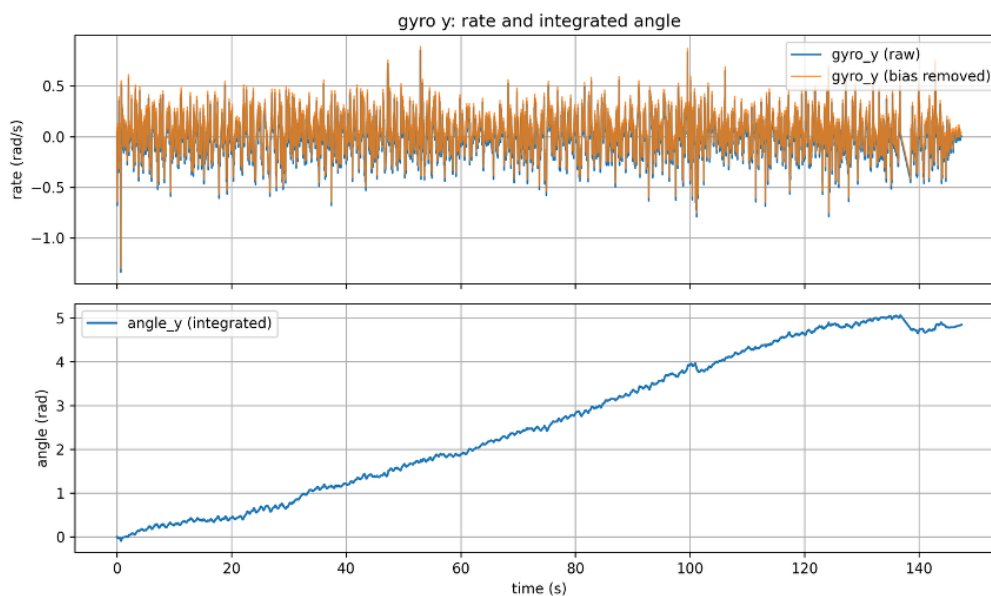


Figure 3: Gyroscope Y-axis: rotational rate and integrated rotation

### 1.4 Gyroscope Z-axis and Magnetometer Heading

**Task:** Plot rotational rate around Z, rotation from the gyro around Z (integrate once), and magnetometer heading (calculate  $\text{atan2}(X/Y)$ ) on three subplots (Fig. 4).

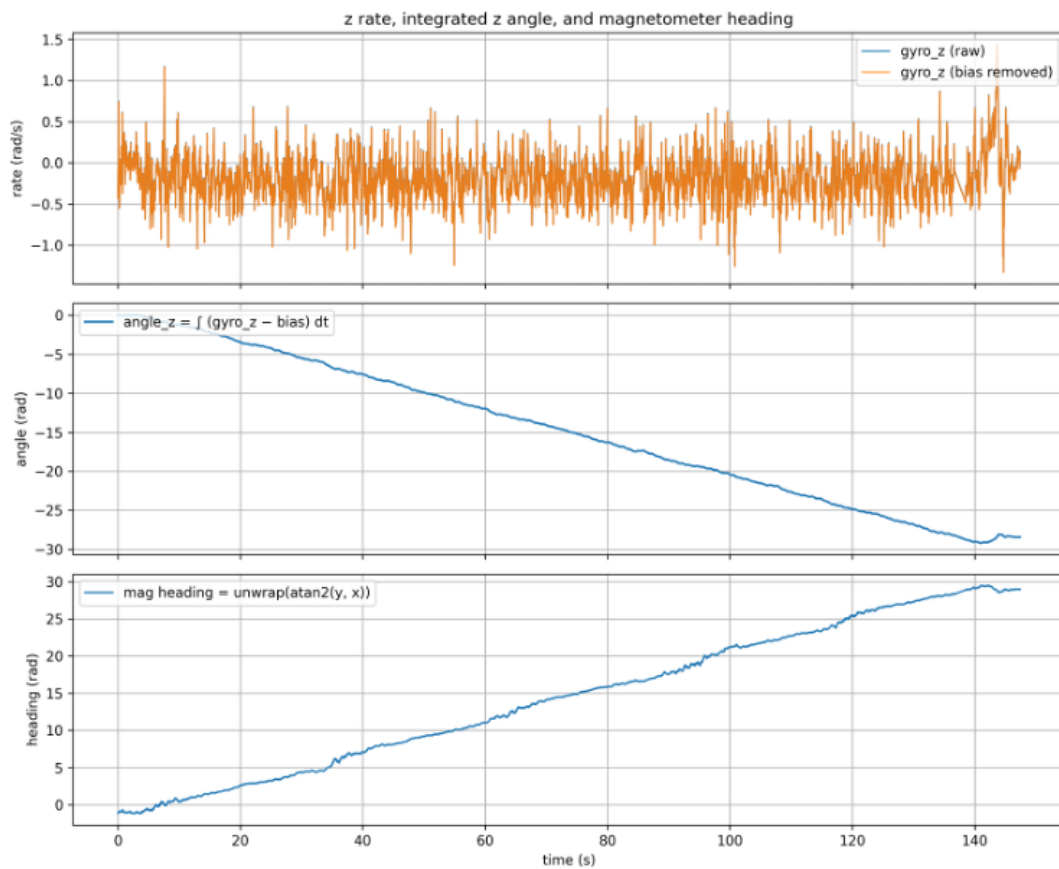


Figure 4: Gyroscope Z-axis rotation and magnetometer heading

### 1.5 Accelerometer X-axis Integration

**Task:** Plot acceleration in X and velocity in X (integrate once) vs. time as two subplots (Fig. 5).

### 1.6 Accelerometer Y-axis Integration

**Task:** Plot acceleration in Y and velocity in Y (integrate once) vs. time as two subplots (Fig. 6).

### 1.7 Accelerometer Z-axis Integration

**Task:** Plot acceleration in Z and velocity in Z (integrate once) vs. time as two subplots (Fig. 7).

### 1.8 Position Reconstruction

**Task:** Using estimates for heading from the magnetometer and the gyro, along with your twice-integrated acceleration in X, make a plot of your N vs. E position (Fig. 8) using different marker types or colors for the magnetometer heading and the gyro heading.

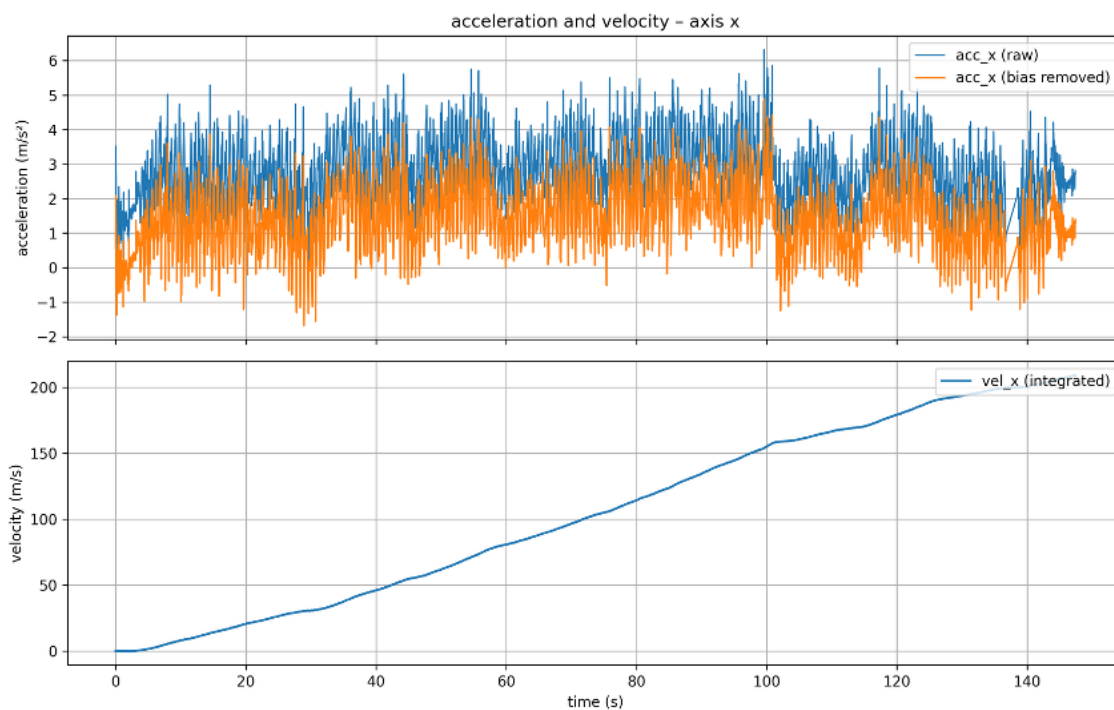


Figure 5: Acceleration and velocity in X-axis

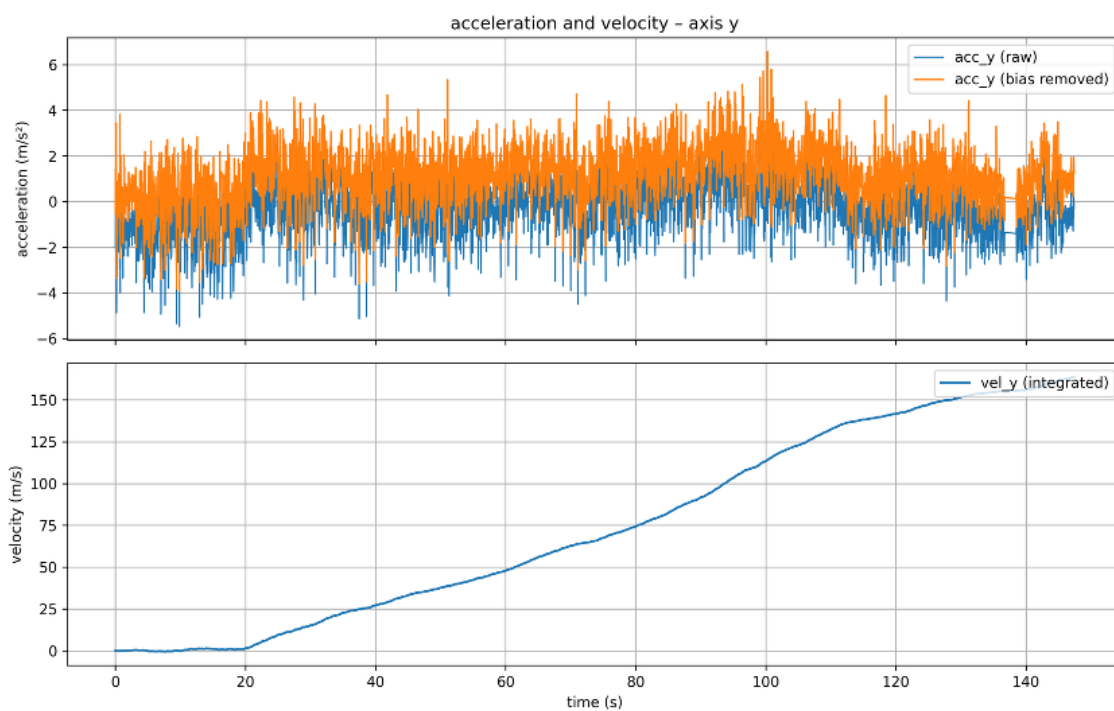


Figure 6: Acceleration and velocity in Y-axis

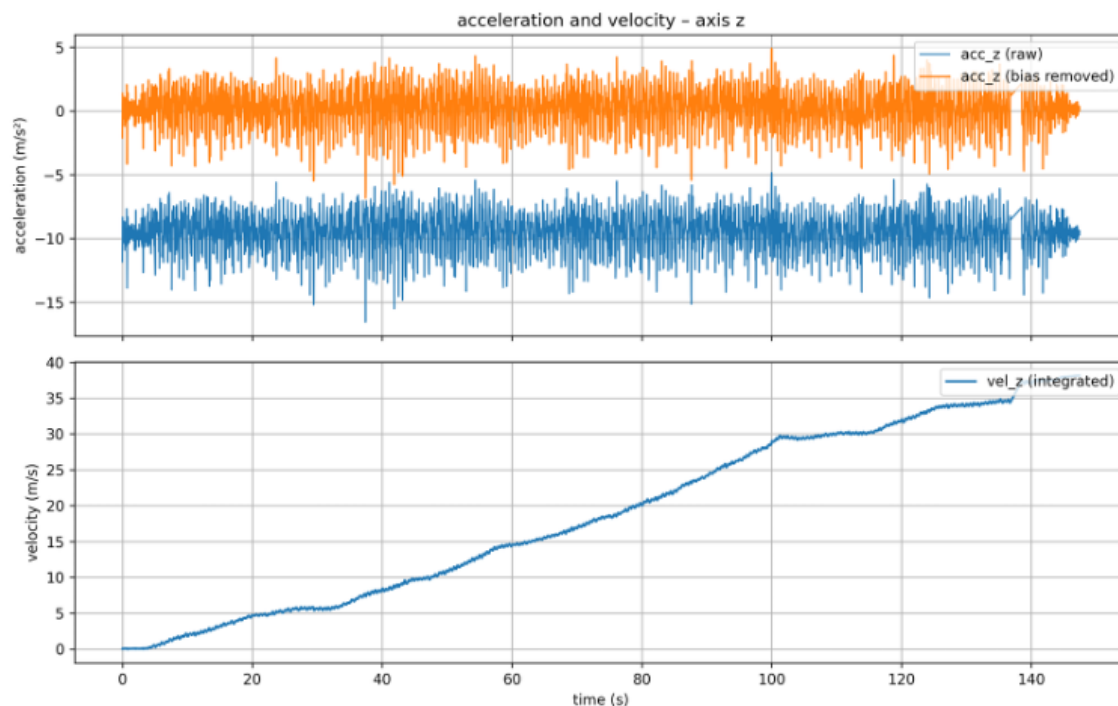


Figure 7: Acceleration and velocity in Z-axis

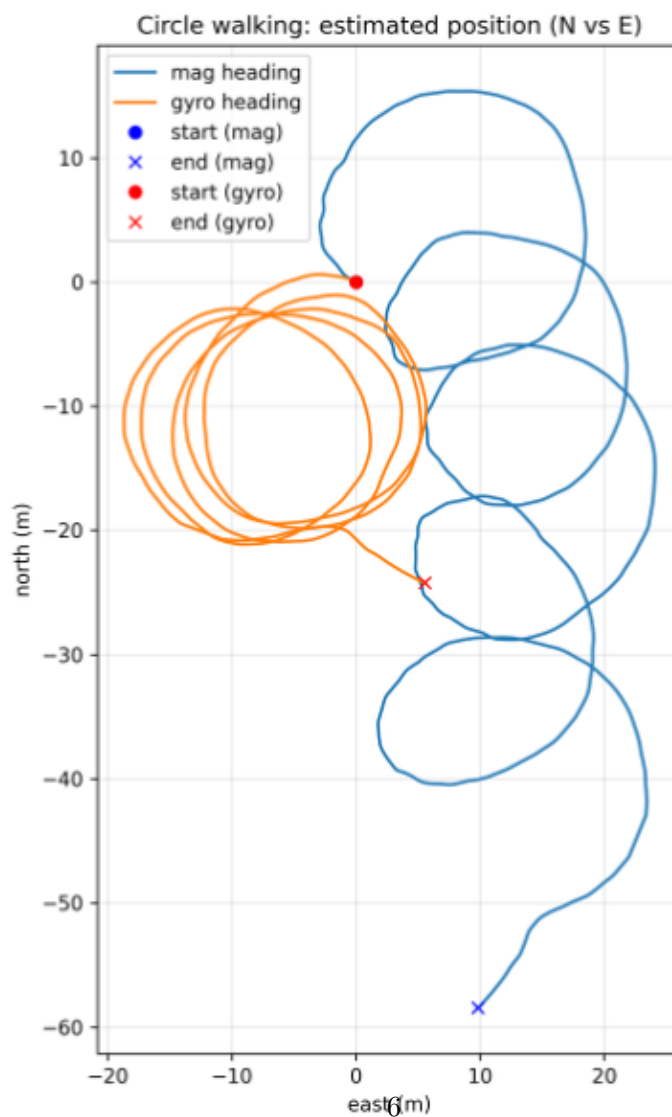


Figure 8: Reconstructed trajectory: Northing vs. Easting position using magnetometer and gyro headings

## 2 Square Walking Data

### 2.1 Magnetometer Calibration

**Task:** Plot the N vs. E components of magnetic field and apply your circle walking calibration to your dataset. Plot two marker styles on the same figure to display the data before/after calibration (Fig. 9).

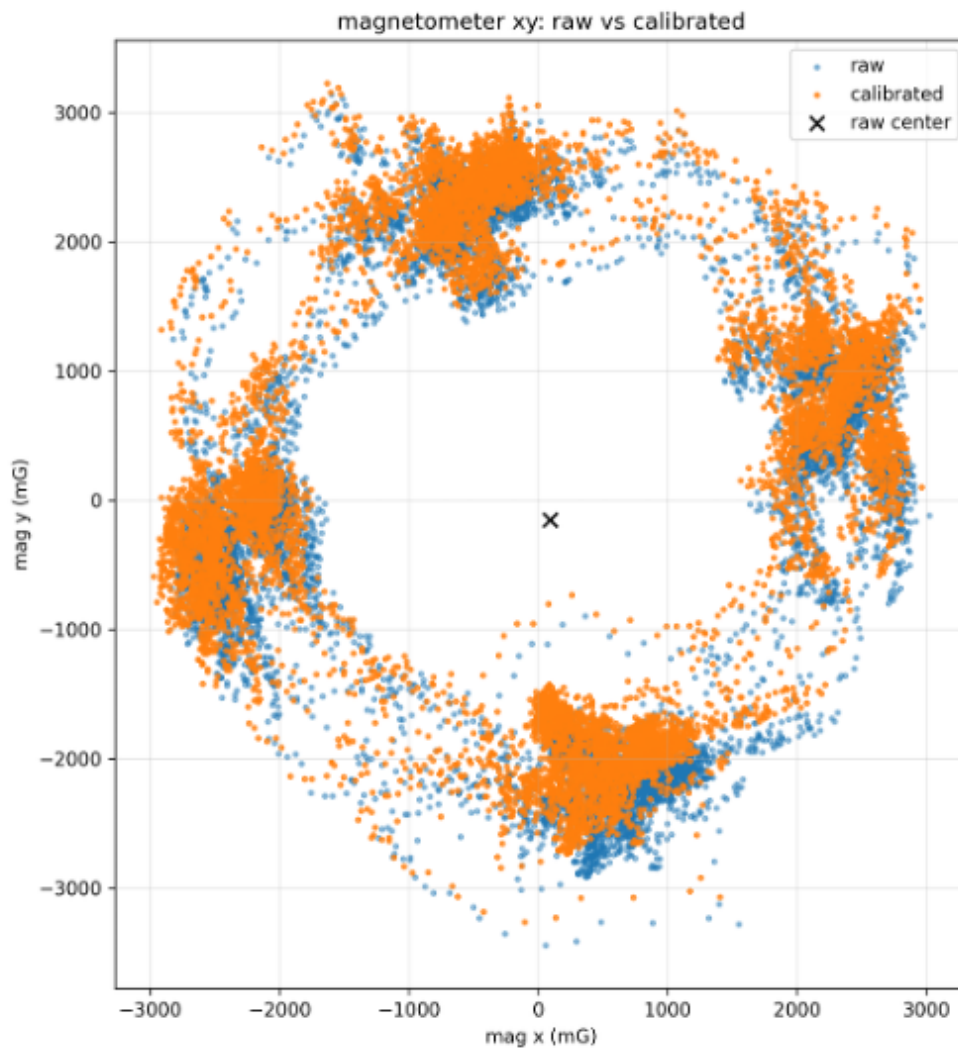


Figure 9: Magnetometer calibration for square walking - before and after calibration

### 2.2 Gyroscope X-axis Integration

**Task:** Plot rotational rate around X and rotation from the gyro around X (integrate once) on two subplots (Fig. 10).

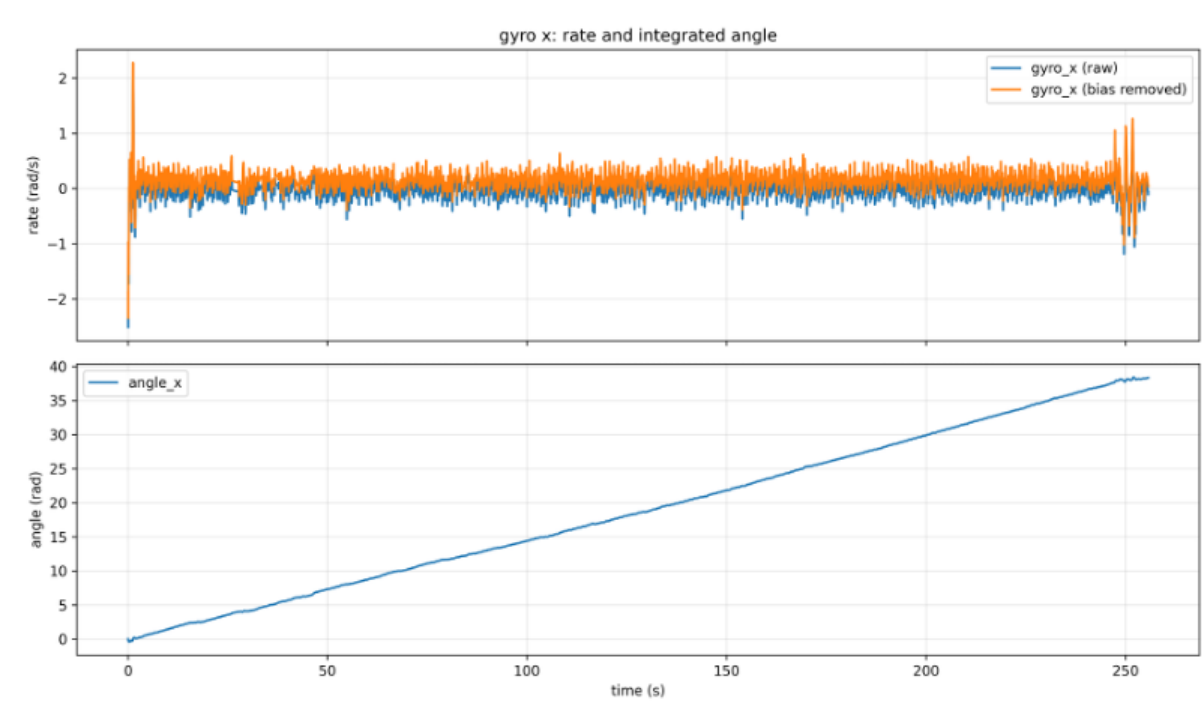


Figure 10: Gyroscope X-axis: rotational rate and integrated rotation

## 2.3 Gyroscope Y-axis Integration

**Task:** Plot rotational rate around Y and rotation from the gyro around Y (integrate once) on two subplots (Fig. 11).

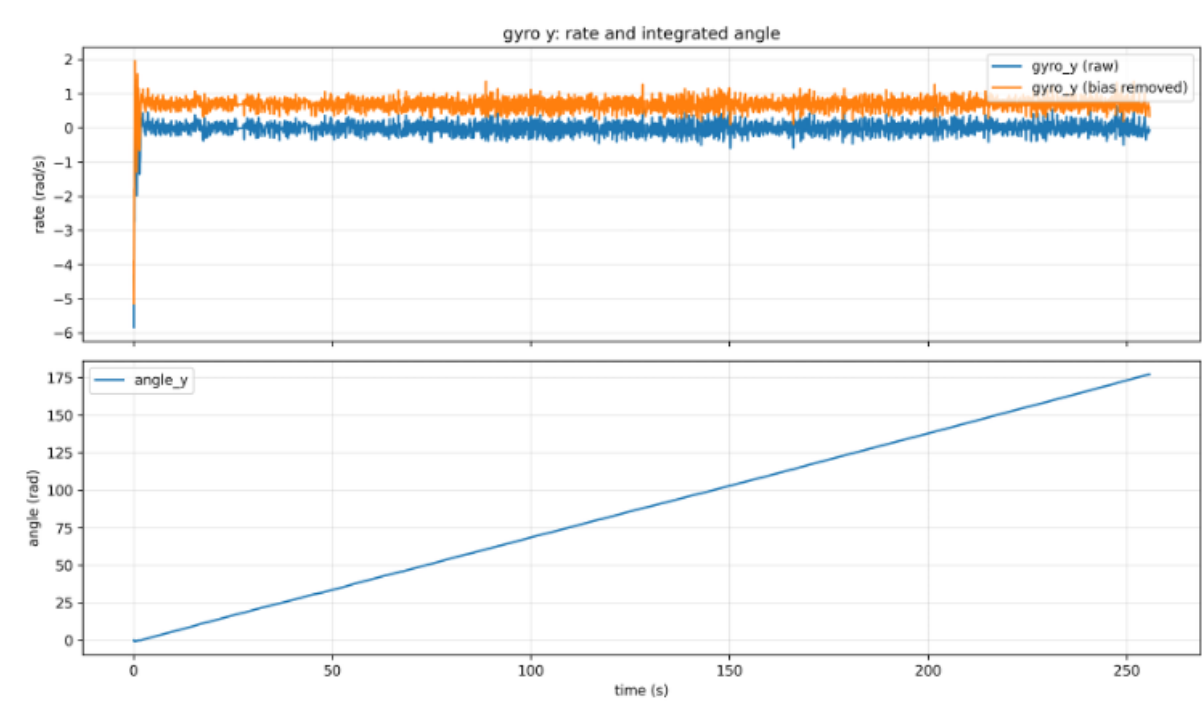


Figure 11: Gyroscope Y-axis: rotational rate and integrated rotation



## 2.4 Gyroscope Z-axis and Magnetometer Heading

**Task:** Plot rotational rate around Z, rotation from the gyro around Z (integrate once), and magnetometer heading (calculate  $\text{atan2}(X/Y)$ ) on three subplots (Fig. 12).

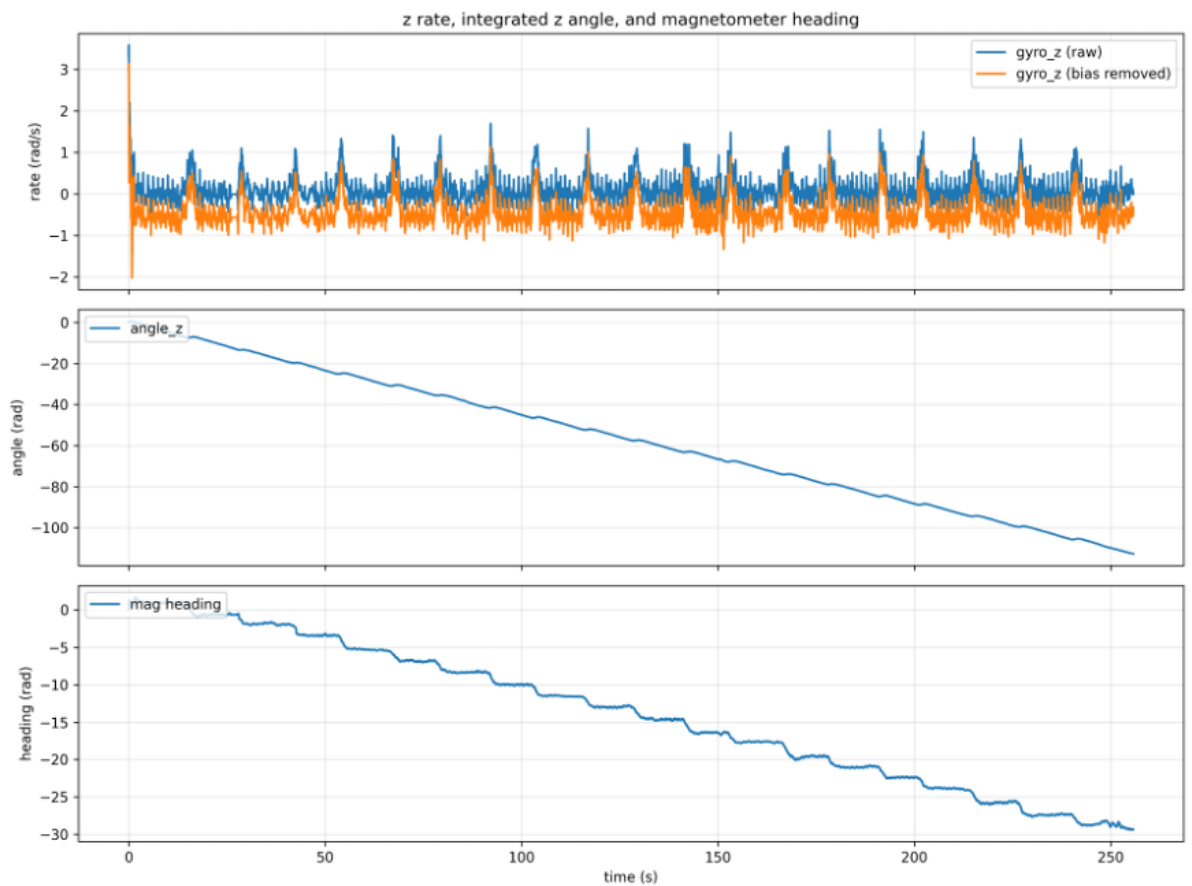


Figure 12: Gyroscope Z-axis rotation and magnetometer heading

## 2.5 Accelerometer X-axis Integration

**Task:** Plot acceleration in X and velocity in X (integrate once) vs. time as two subplots (Fig. 13).

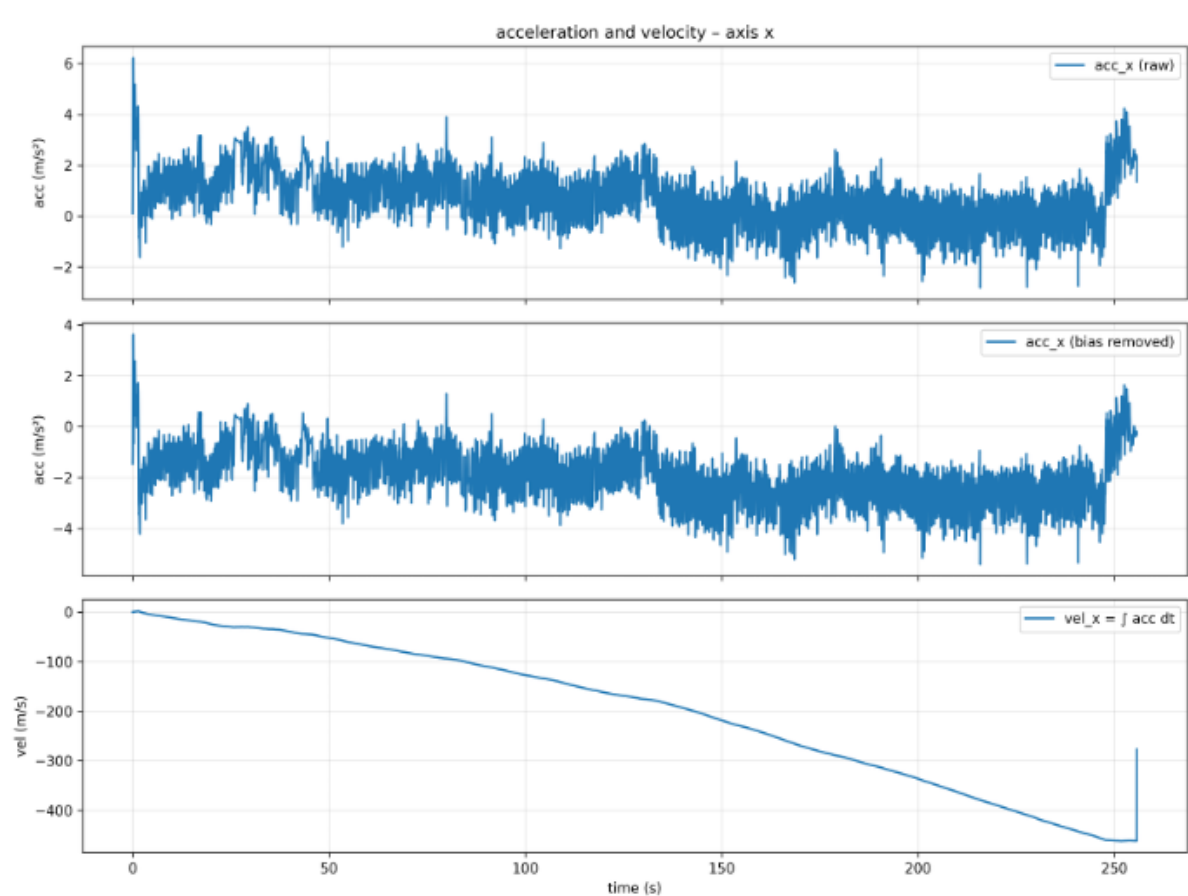


Figure 13: Acceleration and velocity in X-axis

## 2.6 Accelerometer Y-axis Integration

**Task:** Plot acceleration in Y and velocity in Y (integrate once) vs. time as two subplots (Fig. 14).

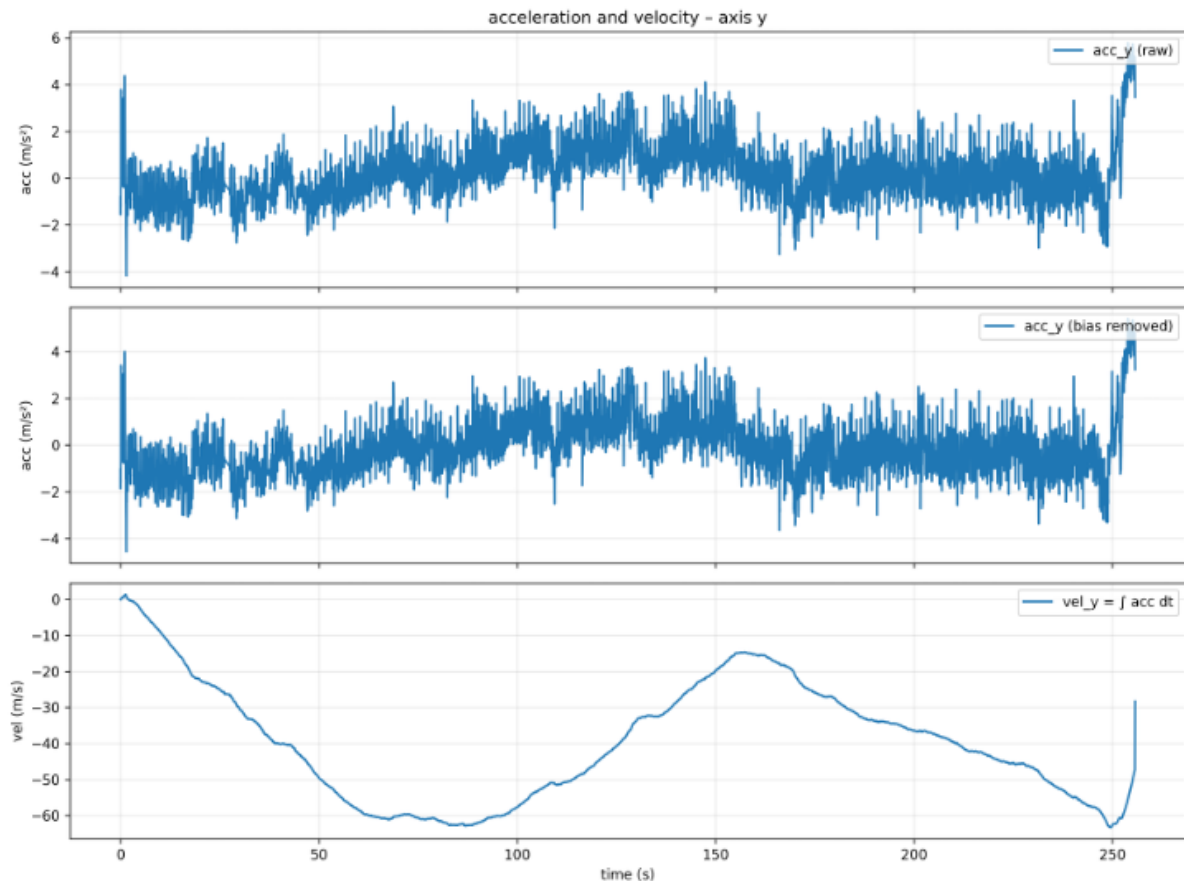


Figure 14: Acceleration and velocity in Y-axis

## 2.7 Accelerometer Z-axis Integration

**Task:** Plot acceleration in Z and velocity in Z (integrate once) vs. time as two subplots (Fig. 15).

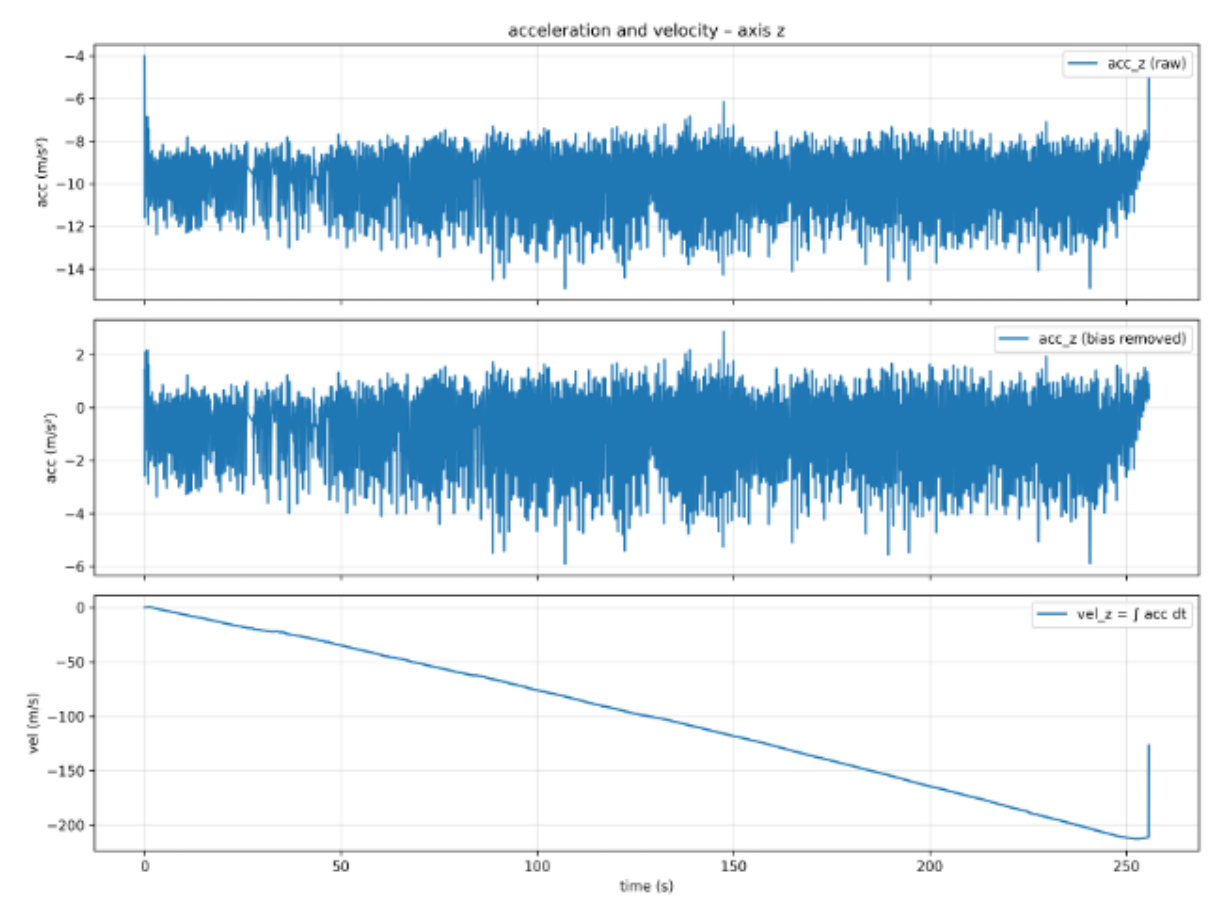


Figure 15: Acceleration and velocity in Z-axis

## 2.8 Position Reconstruction

**Task:** Using your estimates for heading from the magnetometer and the gyro, along with your twice integrated acceleration in X, make a plot of your N vs. E position (Fig. 16) using different marker types or colors for the magnetometer heading and the gyro heading.

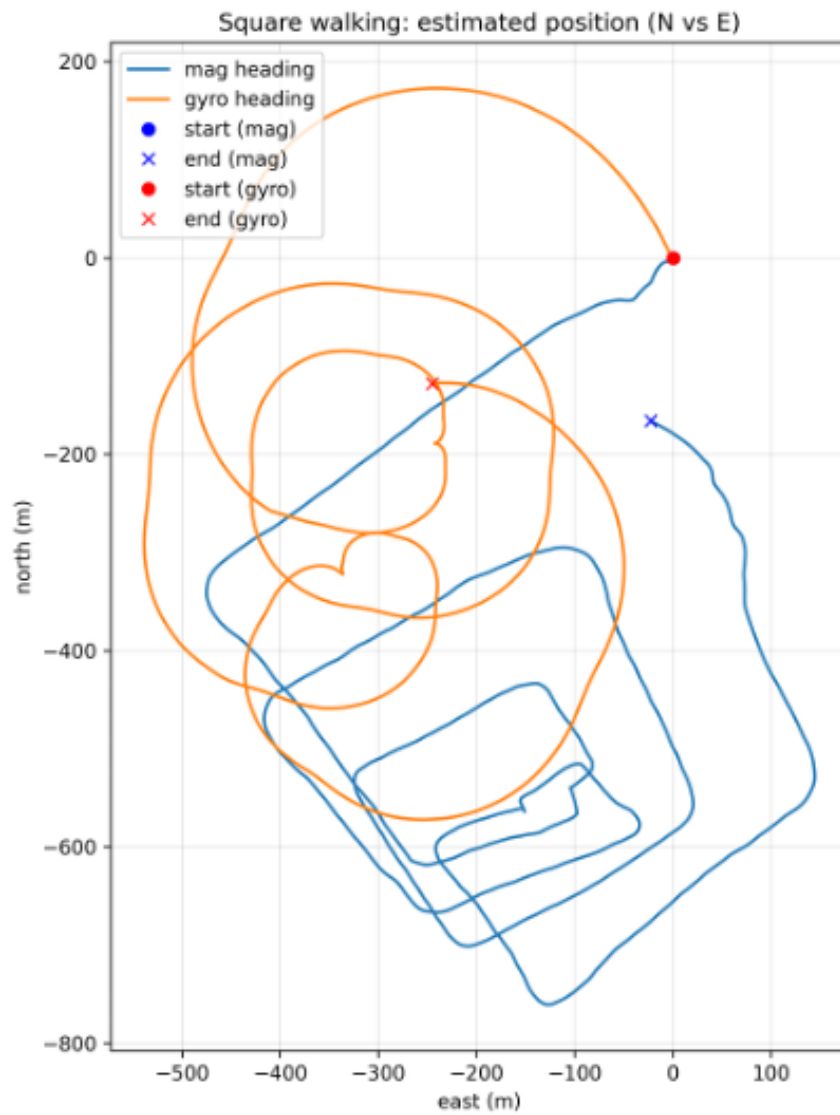


Figure 16: Reconstructed trajectory: Northing vs. Easting position using magnetometer and gyro headings

### 3 Questions and Answers

**Question 1: Write 2-3 sentences about your observations and whether your results "make sense" given the ground truth recording of your video.**

**Answer:**

Across both datasets, we notice some interesting observations.

The raw magnetometer data shows a clearly distorted ellipse, indicating hard-iron and soft-iron errors, likely from nearby metal objects, my phone, laptop or the IMU housing itself.

After applying calibration to the data, the cloud becomes far more circular and recenters to the origin, showing the calibration method is effective. The gyro X/Y/Z rate plots all show low-frequency drift but relatively stable bias (which is why the integrated angles climb linearly over time).

This drift accumulates significantly, demonstrating expected IMU behavior where even tiny bias errors lead to large integrated orientation drift over several minutes. The acceleration to velocity integrations also compounds over time with velocities reaching over 100 m/s, which is expected because we are integrating noisy human motions with no external constraints.

Finally, the Northing-Easting reconstructed trajectory spirals outward and drifts badly compared to the true  $\sim 5$  m trajectory. All of these behaviors are expected for unfiltered and raw data without much filtering from inertial sensors and are in line with what we expect.

**Question 2: Write 2-3 sentences about what revisions you might make to your setup to collect the highest quality data possible for lab 5, when you will be driving around in a car.**

**Answer:**

For the Lab 5 dead reckoning, I expect to mount the IMU as perfectly as possible close to the center of mass and its X-axis aligned with the axis of the car's length. After this, I intend to perform a calibration 'test drive', where we drive in a circle around the Ruggles roundabout.

In the processing front, we'll have to make extensive use of bias correction methods, drift correction methods, high-pass/low-pass filters, complementary filters etc. unlike Lab 4 (here), where we mostly keep raw data for visualizing issues like drift/bias etc.

The data we'll collect include visual-odometry via a camera for our reference, our IMU data, and GPS data for comparison to dead reckoning results.